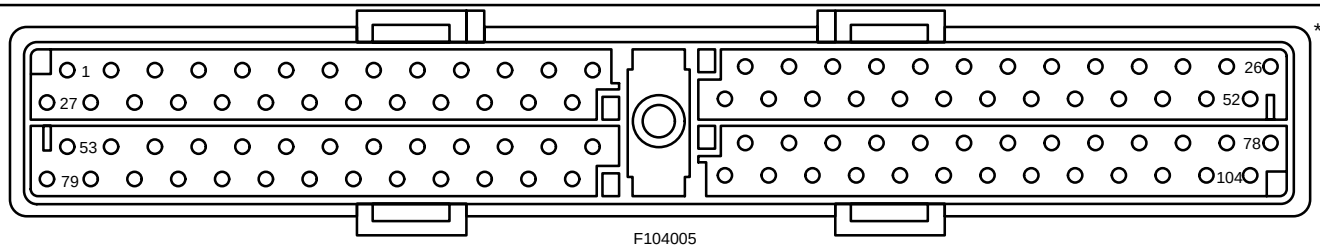
C175 (GY)

12A581

A147
Powertrain Control
Module (PCM)
(12A650)



F104005

*5.4L supercharged

Pin	Circuit	Circuit function
1	315 (VT/OG)	Shift solenoid B
	1025 (OG/YE)	Coil On Plug (COP) 6 (12029), control
2	–	not used
3	1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	570 (BK/WH)	Ground
4	1496 (PK)	Sensor, Transfer case
5	–	not used
6	237 (OG/YE)	Shift solenoid A
7	–	not used
8	190 (WH/OG)	Intake Manifold Runner Control (IMRC) module
9	–	not used
10	–	not used
11	315 (VT/OG)	Shift solenoid B
12	911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
13	107 (VT)	Module programming signal
14	784 (LB/BK)	Four–wheel drive, indicator, Low
15	915 (PK/LB)	SCP Bus –
16	914 (TN/OG)	SCP Bus +
17	–	not used
18	–	not used
19	27 (OG/LG)	Fuel pump high/low relay, control
20	924 (BN/OG)	Powertrain Control Module (PCM) (12A650), to transmission coast clutch solenoid
21	349 (DB)	Crankshaft position sensor (6C315) –
22	350 (GY)	Crankshaft position sensor (6C315) +
23	–	not used
24	570 (BK/WH)	Ground
25	567 (LB/YE)	Ground
	48	Shield

Pin	Circuit	Circuit function
26	526 (DB/LG)	PCM, to, Ignition coil (12029)
	1024 (LG/WH)	Coil On Plug (COP) 1 (12029), control
27	237 (OG/YE)	Shift solenoid A
	1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
28	–	not used
29	224 (TN/WH)	PCM, to transmission overdrive cancel switch
30	–	not used
31	–	not used
32	311 (DG/VT)	Knock sensor (12A699)
33	–	not used
34	1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
35	392 (RD/LG)	Heated Oxygen Sensor (HO2S) #12 (9G444), signal
36	968 (TN/LB)	Mass Air Flow (MAF) sensor (12B579), signal, return
37	923 (OG/BK)	Transmission Fluid Temperature (TFT) sensor, input
38	1291 (RD/YE)	Intake air temperature sensor, input
39	743 (GY)	Intake Air Temperature (IAT) sensor (12A697), signal
40	238 (DG/YE)	Fuel pump relay, switched power, output
	*786 (RD)	Crash fuel shutoff relay, control
41	347 (BK/YE)	A/C clutch relay, switched output
42	367 (BN)	Intake Manifold Runner Control (IMRC) module
	*965 (LG/VT)	Supercharger bypass solenoid, input
43	369 (WH/OG)	Charge air cooler pump relay, control
44	–	not used
45	–	not used
46	367 (BN)	Intake Manifold Runner Control (IMRC) module
47	360 (BN/PK)	EGR Vacuum Regulator (EVR), control
48	–	not used
49	1145 (LB/BK)	Digital Transmission Range (DTR) sensor, TR2
50	1143 (WH/BK)	Digital Transmission Range (DTR) sensor, TR4

Pin	Circuit	Circuit function
51	570 (BK/WH)	Ground
52	527 (RD/LB)	PCM, to, Ignition coil (12029)
	1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
53	1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
54	480 (VT/YE)	Torque Converter Clutch (TCC) solenoid
55	729 (RD/WH)	Voltage supplied at all times (overload protected)
56	191 (LG/BK)	Powertrain Control Module (PCM) (12A650), EVAP canister purge valve
57	310 (YE/RD)	Knock sensor (12A699)
58	–	not used
59	1496 (PK)	Sensor, Transfer case
	970 (DG/WH)	Turbine Shaft Speed (TSS) sensor
60	74 (GY/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472), signal
61	393 (VT/LG)	Heated Oxygen Sensor (HO2S) #22 (9G444), signal
62	791 (RD/PK)	Fuel tank pressure transducer sensor (9C968)
	1164 (LB)	Engine controls, Gas temperature sensor (CNG), Temperature
63	*356 (DB/LG)	Barometric Absolute Pressure (BAP) sensor, monitor
	141 (RD/PK)	Injector pressure sensor
64	199 (LB/YE)	Digital Transmission Range (DTR) sensor (7F293), TR3A
65	352 (BN/LG)	Differential Pressure Feedback EGR (DPFE) sensor, input
	48	not used
66	1102 (YE/LG)	Cylinder–head temperature sensor (6G004), to, Powertrain Control Module (PCM) (12A650)
67	91 (VT/WH)	Canister vent
68	679 (GY/BK)	Vehicle Speed Sensor (VSS) +
69	331 (PK/YE)	A/C clutch relay, control
70	1017 (BK/WH)	Check cap indicator lamp
71	361 (RD)	Voltage supplied in Start and Run (overload protected)
72	561 (TN/RD)	Fuel injector 7, control
	1171 (WH)	Fuel injector 7, control
73	559 (TN/BK)	Fuel injector 5, control
	1169 (WH/LB)	Fuel injector 5, control
74	557 (BN/YE)	Fuel injector 3, control
	1167 (WH/BK)	Fuel injector 3, control

Pin	Circuit	Circuit function
75	555 (TN)	Fuel injector 1, control
	1165 (WH/RD)	Fuel injector 1, control
76	570 (BK/WH)	Ground
77	570 (BK/WH)	Ground
78	528 (PK/WH)	PCM, to, Ignition coil (12029)
	1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
79	911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	1029 (WH/RD)	Coil On Plug (COP) 8 (12029), control
80	926 (LB/OG)	Fuel pump relay, control
81	925 (WH/YE)	Electronic Pressure Control (EPC) solenoid, input
82	–	not used
83	264 (WH/LB)	Idle Air Control (IAC) valve (9F715), control
84	136 (DB/YE)	Powertrain Control Module (PCM) (12A650) – Output Shaft Speed (OSS) sensor (7M101)
85	795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
86	–	not used
87	94 (RD/BK)	Heated Oxygen Sensor (HO2S) #21 (9F472), signal
88	967 (LB/RD)	Mass Air Flow (MAF) sensor (12B579), output
89	355 (GY/WH)	Throttle Position Sensor (TPS) (9B989), input
90	351 (BN/WH)	Reference, voltage
91	359 (GY/RD)	signal, return
92	810 (RD/LG)	Brake pedal position switch (13480), input
93	387 (RD/WH)	Heated Oxygen Sensor (HO2S) #11 (9F472), heater
94	388 (YE/LB)	Heated Oxygen Sensor (HO2S) #21 (9F472)
95	389 (WH/BK)	Heated Oxygen Sensor (HO2S) #12 (9G444)
96	390 (TN/YE)	Heated Oxygen Sensor (HO2S) #22 (9G444), heater
97	361 (RD)	Voltage supplied in Start and Run (overload protected)
98	562 (LB)	Fuel injector 8, control
	1172 (YE)	Fuel injector 8, control
99	560 (LG/OG)	Fuel injector 6, control
	1170 (YE/LB)	Fuel injector 6, control
100	558 (BN/LB)	Fuel injector 4, control
	1168 (YE/BK)	Fuel injector 4, control
101	556 (WH)	Fuel injector 2, control
	1166 (YE/RD)	Fuel injector 2, control

Pin	Circuit	Circuit function
102	–	not used
103	570 (BK/WH)	Ground
104	1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control